

Retail Sales Analysis - Bindhu Gadham

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Retail Sales Analysis

End-to-end sales & profitability analysis of 9,800 retail order line items, with a normalized database, a full SQL analysis layer, and an interactive dashboard.

Executive summary

This project analyzes 9,800 retail order line items (2020–2023) to evaluate sales, profitability, customer behavior, product performance, and regional trends, using a normalized MySQL database and a SQL-based reporting layer feeding an interactive dashboard.

\$23.3Mtotal sales

\$2.63Mtotal profit

11.3%profit margin

3,895orders

776customers

The most actionable finding is a clean, quantified relationship between discounting and profitability: margin holds around 23% with no discount, turns negative in the 16–30% discount band, and collapses to roughly -68% at 50%+ discount. Furniture — specifically Tables — is the one major sub-category losing money overall (-11.7% margin on \$2.3M in sales), worth a pricing or discount-cap policy conversation on its own.

A note on this environment. The sandbox this was built in has no root access and no network. MySQL server isn't installable without root/apt, so the SQL here (schema, load script, 10 analysis queries, reporting views) is real MySQL syntax meant to run against an actual MySQL instance — validated in this environment by running the equivalent query logic through Python's built-in SQLite against the same generated dataset. Every number in this report comes from an actually-executed query, not an assumption. Power BI Desktop is Windows-only and can't run here at all, so an interactive HTML dashboard was built covering the same ground, plus a written guide with the exact DAX measures to rebuild it in real Power BI Desktop.

Problem statement

A retail business needs to know where it's actually making money — which categories, regions, and customer segments drive profit rather than just revenue — and whether common practices like

discounting are helping or quietly eroding margin. This project builds the database and analysis layer to answer that with real queries rather than intuition.

Data & methodology

Dataset: 9,800 simulated order line items across 4 US regions, 3 customer segments (Consumer, Corporate, Home Office), and 3 product categories (Furniture, Office Supplies, Technology, with their sub-categories), spanning 2020–2023. Styled after the well-known "Sample Superstore" dataset, including its best-documented finding (heavy discounts erode Furniture margin) reproduced here from first principles rather than copied.

Database design: rather than analyzing one flat CSV in place, the data is normalized into customers / products / orders / order_details tables with primary and foreign keys and indexes on the columns the analysis queries filter and join on (order date, customer ID, product ID, region). A staging table mirrors the raw CSV for LOAD DATA INFILE, then INSERT...SELECT populates the normalized schema.

SQL analysis layer: 10 queries answer the core business questions — an executive KPI summary, category/sub-category performance, top products (and products losing money), regional and state performance, customer segment analysis, top customers by lifetime value, monthly trend with year-over-year growth (self-join), discount-band profit margin analysis, shipping mode mix, and a running cumulative sales total using window functions (SUM() OVER, RANK() OVER). A views layer wraps the fact table and these aggregations into reporting views, so the semantic layer — what "profit margin" means, how region rolls up — lives in the database once rather than being redefined in every BI tool that connects to it.

Key results

Fig. 1 — Technology drives most of both sales and profit; Furniture is nearly break-even overall once Tables' losses are netted against Chairs/Furnishings/Bookcases.

Fig. 2 — The single most useful chart in the analysis: profit margin crosses zero between the 1–15% and 16–30% discount bands, and keeps falling from there. The business is effectively subsidizing large discounts out of margin, not just revenue.

Fig. 3 — East leads in total sales; margin is fairly consistent across regions (10.8%–11.8%), so the discounting issue above is a company-wide pattern, not a regional one.

Fig. 4 — Clear November/December holiday seasonality every year, with an overall upward trend from 2020 into 2023 (year-over-year growth: -6.8% in 2021, +7.2% in 2022, -4.4% in 2023).

Fig. 5 — Consumer is the largest segment by sales share, but Corporate has the highest sales-per-customer (\$31,350 vs. \$30,202 for Consumer and \$27,500 for Home Office).

Business recommendations

Cap discounts at or below ~15% as a default policy — margin is still positive through that band and turns negative beyond it across the whole dataset, not just in one category.

Review Tables and Bookcases (Furniture) pricing specifically — both lose money overall, and Tables alone accounts for -\$269,707 in profit against \$2.3M in sales; either the cost basis, list price, or discount practice on this sub-category needs to change.

Double down on Technology, especially Copiers (20.2% margin, largest profit contributor) — the highest-revenue and highest-margin category, an unusually good combination worth protecting with inventory/marketing priority.

Investigate Corporate segment growth — highest sales-per-customer despite being the second-largest segment by customer count, suggesting room to grow via acquisition rather than just deepening existing accounts.

Plan inventory and staffing around the consistent November/December demand spike visible in every year of data.

Interactive dashboard

A self-contained, filterable dashboard (Year / Region / Category / Segment) built on the same line-item data as the SQL analysis — KPI cards, a monthly sales/profit trend, category and regional breakdowns, the discount-vs-margin chart, a segment donut, and a live top-10-products table, all recomputed in the browser as filters change. A companion Power BI guide gives the exact DAX measures and visual-by-visual steps to rebuild the same report in Power BI Desktop against the MySQL database.

Tech stack

MySQL

SQL (window functions, views)

Python

Pandas

SQLite (validation)

Chart.js

Power BI / DAX

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